Recursion as a Load-Bearing Principle: A Synthesis for the VRU / DPPU Architecture

TL;DR
- **Recursion is not one technique among many — it is the single load-bearing primitive that unifies computation, geometry, compression, and cognition.** Fixed-point structure (Banach / Kleene / Tarski), self-similarity (fractals, renormalization-group flows), and bounded-state sequence modeling (RNN ↔ linear attention ↔ SSM) are the same mathematical object viewed through different lenses, and Katharopoulos et al. (ICML 2020) followed by Dao & Gu (2024, "Structured State Space Duality") have made the equivalence formal.
- **The recurrent comeback is real and technically grounded, but narrowly bounded.** Mamba/Mamba-2, xLSTM, Griffin/Hawk, RWKV-v5/6, and Mixture-of-Recursions (Bae et al., arXiv:2507.10524, July 2025) match or beat Transformer baselines on language perplexity at sub-7B scale with linear time and bounded-state inference, but they remain weaker on exact-retrieval and on certain in-context-learning tasks — Park et al. (arXiv:2402.04248) showed Mamba (tested up to 2.8B) "fails in vector-valued MQAR" [arXiv](https://arxiv.org/pdf/2402.04248v1) while matching Transformers on standard regression ICL. VRU/DPPU lives or dies on whether φ = 4/π gives a *derivable* (not cosmetic) geometric scaling that bends the parameter–capability frontier.
- **4/π is a real, recurring mathematical constant with a single root cause.** It is the angular average $\tfrac{2}{\pi}\!\int_0^{\pi}\!|\sin\theta|\,d\theta = 4/\pi$, which is exactly the amplitude of the fundamental Fourier harmonic of a unit square wave, the ratio of a square to its inscribed circle, the value of Brouncker's 1655 continued fraction, the cube's mean square width minus one (Finch 2016), and a doubling of Buffon's $2/\pi$. It is *not* sharp in the plane isoperimetric inequality (4π is), in Wigner/Marchenko–Pastur (2π is), or in sinusoid form factor (π/(2√2) is). Whether 4/π is principled or decorative in VRU depends entirely on whether the recurrence's state update can be derived as such an angular average.

Key Findings

1. **Recursion has a unified mathematical core: the fixed point.** Every domain in scope reduces to one of three patterns: (a) a fixed point of an operator (Banach, Brouncker, Kleene, Knaster–Tarski, Y-combinator, deep equilibrium models, RG fixed points, predictive-coding attractors); (b) an initial algebra / catamorphism over an inductively defined structure (lists, trees, syntactic recursion); (c) a coalgebra / corecursion generating an infinite stream (RNN dynamics, Markov processes, Neural ODE trajectories). The Curry–Howard correspondence collapses (a) and (b) into the same object: a recursive program *is* a proof by induction on a well-founded structure.

2. **The Transformer's victory over RNNs was a hardware and gradient story, not a representational one.** Three reasons it happened: vanishing/exploding gradients in BPTT (Hochreiter 1991; Bengio, Simard, Frasconi 1994); inability to parallelize across the sequence dimension during training (sequential dependency in $h_t = f(h_{t-1}, x_t)$); and softmax attention's content-based mixing being qualitatively more expressive for retrieval and induction-head behavior. Katharopoulos et al. (arXiv:2006.16236) proved the converse formally:

a transformer with kernel feature map ϕ and no softmax is *exactly* an RNN with matrix state $S_t = S_{t-1} + \phi(k_t)v_t^\top$. [arxiv](https://arxiv.org/pdf/2006.16236) Their stated empirical claim: "Our linear transformers achieve similar performance to vanilla transformers and they are up to 4000x faster on autoregressive prediction of very long sequences." [arXiv](https://arxiv.org/abs/2006.16236) [arxiv](https://arxiv.org/pdf/2006.16236) Dao & Gu (ICML 2024) extended this with Structured State Space Duality (SSD), showing Mamba-2 and a class of masked attention are the same operator under different computational schedules.

3. **The recurrent renaissance is concentrated in five families:** (i) **Selective SSMs** — HiPPO → S4 → Mamba → Mamba-2 → Mamba-3 (Gu, Dao, Ré et al., 2020–2026); (ii) **Linear attention / gated RNNs** — Linear Transformer, RWKV v4–v6, RetNet, GLA, Hawk/Griffin (DeepMind, arXiv:2402.19427); (iii) **Extended LSTMs** — xLSTM (Beck, Pöppel, Hochreiter et al., NeurIPS 2024 spotlight) and xLSTM-7B (2025); (iv) **Recursive/looped Transformers** — Universal Transformer (Dehghani et al., ICLR 2019), Relaxed Recursive Transformers (Bae et al. 2024), Mixture-of-Recursions (Bae et al., arXiv:2507.10524); (v) **Implicit/continuous models** — Neural ODEs (Chen, Rubanova, Bettencourt, Duvenaud, NeurIPS 2018) and Deep Equilibrium Models (Bai, Kolter, Koltun, NeurIPS 2019).

4. **The strongest single empirical result for recurrence at scale is Griffin.** Soham De et al. (DeepMind, arXiv:2402.19427) report: "Hawk exceeds the reported performance of Mamba on downstream tasks, while Griffin matches the performance of Llama-2 despite being trained on over 6 times fewer tokens." [arXiv](https://arxiv.org/abs/2402.19427) Griffin shows power-law scaling of held-out loss in training FLOPs up to 14B parameters, extrapolates beyond training sequence length, [arXiv](https://arxiv.org/html/2402.19427v1) and has lower inference latency / higher throughput than an MQA Transformer baseline especially at long context.

5. **The strongest arguments *against* a recurrent comeback at frontier scale.** (a) On retrieval-style tasks, pure SSMs lose: Park et al. (arXiv:2402.04248) found Mamba (up to 2.8B) "fails in vector-valued MQAR" though it matches Transformers on standard regression ICL. (b) Hybrid recipes (Griffin, Jamba, Zamba) win by adding back local attention; the pure-recurrence camp has not yet shipped a frontier-quality model >70B. (c) Sparse/native attention (DeepSeek NSA 2025, MoBA 2025) is attacking the KV-cache problem from the opposite direction without giving up softmax. (d) Compute-optimal scaling laws (Hoffmann et al. 2022, *Chinchilla*) were measured on Transformers and the optimal token-to-parameter ratio for SSMs is unsettled; MosaicML's Sardana et al. (arXiv:2401.00448) showed "model quality continues to improve as we scale tokens per parameter to extreme ranges (up to 10,000)" [arxiv](https://arxiv.org/html/2401.00448v2) for inference-cost-aware deployment.

6. **4/π is a sharp, principled constant in seven well-attested places.** (i) The fundamental Fourier harmonic of a unit-amplitude square wave has amplitude exactly $4/\pi$, from $b_1 = \tfrac{2}{\pi}\!\int_0^\pi\!\sin x\,dx = 4/\pi$ — so $f(x) = \tfrac{4}{\pi}\sum_{k\geq 1}\tfrac{\sin((2k-1)x)}{2k-1}$; [Hmc](https://saeta.physics.hmc.edu/p064/FO-Gibbs.html) (ii) the ratio of a square's area to its inscribed circle's area is exactly $4/\pi$; [Quora](https://www.quora.com/A-circle-is-inscribed-in-a-square-What-is-the-ratio-of-the-area-of

-the-circle-to-the-area-of-the-square) (iii) Brouncker's 1655 continued fraction $\tfrac{4}{\pi} = 1 + \tfrac{1^2}{2 + \tfrac{3^2}{2 + \tfrac{5^2}{2 + \cdots}}}$; [Wikipedia](https://en.wikipedia.org/wiki/William_Brouncker,_2nd_Viscount_Brouncker) (iv) the Wallis product, inverse form; (v) doubled Buffon's needle ($2P = 4/\pi$ for $L = d$); (vi) Steven R. Finch's result that the cube's mean *square* width equals $1 + 4/\pi$ [arxiv](https://arxiv.org/pdf/1110.0671) (Cauchy-mean-width family, arXiv:1110.0671); (vii) Winitzki's elementary erf approximation, where the coefficient $4/\pi$ is forced by the small-x Gaussian asymptotics. In nearly every legitimate case, the constant arises from the angular average $\tfrac{2}{\pi}\!\int_0^\pi |\sin\theta|\,d\theta = 4/\pi$.

7. **Compression and intelligence are formally linked, and recursion is the mechanism by which compression actually happens.** Solomonoff–Kolmogorov–Chaitin algorithmic complexity defines the shortest program that outputs a string; Hutter's AIXI (2005) couples this with sequential decision theory to define a universal — but uncomputable — RL agent; the Hutter Prize for lossless text compression operationalizes this on enwik9. Every short program that generates a long output does so by recursion or iteration. A parameter-efficient model is structurally a model whose program is short relative to what it predicts — and weight-tying / recursion / parameter sharing across depth is the architectural realization of that principle. This is the philosophical grounding for Dylan's intuition: recursion is the compression mechanism.

Details

1. Mathematical foundations of recursion

Recursive function theory. Gödel's primitive-recursive functions, Church's lambda calculus, Turing's machines, and Kleene's μ-recursive functions were proven equivalent by 1936 — the Church–Turing thesis. The crucial divide is between *primitive recursive* (definable by composition and bounded recursion on the predecessor; always terminates) and *general recursive / μ-recursive* (Turing-computable; allows unbounded minimization). Ackermann's function is the canonical total computable function that is not primitive recursive. Kleene's *first recursion theorem* gives the fixed point of every computable functional; the *second recursion theorem* (Kleene 1938) is the formal engine behind the Y combinator: $Y = \lambda f.(\lambda x.f(xx))(\lambda x.f(xx))$, satisfying $Yf = f(Yf)$.

Fixed-point theorems as the substrate.
- *Banach (1922)*: a contraction on a complete metric space has a unique fixed point, found by iteration — the existence/uniqueness lemma behind Picard–Lindelöf for ODEs and convergence of deep equilibrium models.
- *Brouwer (1911)*: every continuous self-map of a compact convex set has a fixed point — non-constructive.
- *Kleene (1952)*: least fixed point of a Scott-continuous functional on a CPO — semantic foundation of recursive definitions in programming languages.

- *Knaster–Tarski (1955)*: every monotone operator on a complete lattice has a complete lattice of fixed points — basis for inductive definitions, abstract interpretation, and monotone-operator DEQs (Winston & Kolter 2020).

Number-theoretic recursion. Continued fractions are recursion incarnate: $[a_0; a_1, a_2, \ldots] = a_0 + 1/(a_1 + 1/(a_2 + \cdots))$. Fibonacci $F_{n+1} = F_n + F_{n-1}$ has Binet form $F_n = (\varphi^n - \psi^n)/\sqrt 5$, and $F_{n+1}/F_n \to \varphi = (1+\sqrt 5)/2$, the unique positive fixed point of $x \mapsto 1 + 1/x$. **Critical caveat for Dylan**: the golden ratio $\varphi \approx 1.618$ and $\phi = 4/\pi \approx 1.273$ are *different kinds of constants*. φ is the fixed point of a recursion; $4/\pi$ is an angular-average integral. Conflating them — or even using the same symbol — would be a category error and a fatal expository move.

Self-similar geometry. Mandelbrot's Hausdorff dimension generalizes integer dimension via covering numbers; for self-similar sets satisfying the open-set condition, $\sum r_i^{d_H} = 1$ (Moran). The Koch curve has $d_H = \log 4/\log 3 \approx 1.262$ — numerically close to $4/\pi \approx 1.273$ but structurally unrelated. Julia and Mandelbrot sets are the fixed-point geometry of iterated polynomials $z \mapsto z^2 + c$.

Category theory and recursion. F-algebras formalize inductive types: an algebra for an endofunctor F is $(A, \alpha: FA \to A)$; the initial algebra $(\mu F, \mathrm{in})$ gives recursion via the unique *catamorphism* fold. Coalgebras dually give *corecursion* (anamorphisms) and infinite structures; *hylomorphisms* compose ana- then cata-, the algebraic skeleton of divide-and-conquer. This is not academic decoration: it is the formal reason a Mamba selective scan can be written as a parallel prefix scan — the recursion has the algebraic structure of a fold over a monoid (associativity is what enables Blelloch-style parallel scan).

Lambda calculus and Y. Lambda calculus expresses recursion *without naming functions* via fixed-point combinators. Recursion *is* finding a fixed point.

2. Computational recursion

Standard taxonomy: *structural recursion* (terminates, recurses on substructure — folds over inductive types); *generative/non-structural recursion* (terminates by a well-founded order — e.g., quicksort); *tail recursion* (compiled to iteration, constant stack); *mutual recursion*. The relation to iteration is exact: every primitive recursive function is a bounded loop; every Turing-computable function an unbounded loop with state. Memoization plus structural recursion equals dynamic programming. Recursive descent parsing realizes context-free grammars; the pumping lemma and Chomsky hierarchy classify languages by the recursive power required. Curry–Howard makes a recursive program literally a proof by induction on a well-founded relation — every recursive call is an appeal to the inductive hypothesis.

3. Recursion in ML and neural architectures (the spine)

Pre-history (1982–2014). Hopfield (1982) — recurrent networks as energy-minimization fixed points. Jordan (1986), Elman (1990) — simple RNNs with hidden-state feedback. Hochreiter & Schmidhuber (1997) — LSTM, with the *constant error carousel* and input/output/forget gates explicitly designed to defeat vanishing gradients. Cho et al. (2014) — GRU as a simplified LSTM. By 2014 LSTMs underpinned Google Translate and Karpathy's char-RNN.

The Transformer takeover (2017–2022). Vaswani et al. (2017): attention plus positional encoding replaces recurrence and parallelizes fully across the sequence. Three reasons RNNs lost: (1) sequential training (no parallelism along t); (2) gradient pathologies despite LSTM gating; (3) attention's content-based mixing is qualitatively more expressive for retrieval. Kaplan et al. (2020) and Hoffmann et al. (2022, *Chinchilla*) established compute-optimal scaling on Transformers: for every doubling of parameters, double the training tokens, [Epoch AI](https://epoch.ai/publications/chinchilla-scaling-a-replication-attempt) empirically ~20 tokens/parameter. [Epoch AI](https://epoch.ai/blog/chinchilla-scaling-a-replication-attempt)

The recurrent renaissance (2020–2025), by lineage.

(a) State-space models. Gu, Dao, Ermon, Rudra, Ré, "HiPPO: Recurrent Memory with Optimal Polynomial Projections" [GitHub](https://github.com/HazyResearch/hippo-code) (NeurIPS 2020) derived a state matrix from projecting input onto Legendre polynomials with a decaying measure, giving provably optimal online function approximation. S4 (Gu, Goel, Ré, ICLR 2022) made this a deep architecture with structured matrices supporting both convolutional and recurrent computation. **Mamba** (Gu & Dao, arXiv:2312.00752) added input-dependent *selectivity* — state-transition parameters become functions of the current token, recovering content-based reasoning that LTI SSMs lacked, with a hardware-aware parallel scan. The paper reports "up to 5x inference throughput improvements in some settings" [ALLPCB](https://www.allpcb.com/allelectrohub/mamba-new-selective-state-space-model-vs-transformer) over comparable Transformers, and that Mamba-3B "outperforms Transformer models of comparable size and is competitive with Transformer models roughly twice its size on language modeling benchmarks." [ALLPCB](https://www.allpcb.com/allelectrohub/mamba-new-selective-state-space-model-vs-transformer) **Mamba-2** (Dao & Gu, ICML 2024) proved the SSD duality — a sub-class of selective SSMs *is* a sub-class of masked attention. Mamba-3 (Lahoti et al., 2026) is the current iteration.

(b) Linear attention / gated RNNs. Linear Transformer (Katharopoulos et al., arXiv:2006.16236) replaces $\mathrm{softmax}(QK^\top)V$ with $\phi(Q)(\phi(K)^\top V)$ plus an RNN-style cumulative state. RetNet (Sun et al., arXiv:2307.08621) and RWKV (Peng et al., arXiv:2305.13048; v5/v6 follow-ups) instantiate this with explicit retention/decay and gating. **Hawk and Griffin** (De et al., arXiv:2402.19427) introduce the real-gated linear recurrent unit (RG-LRU). "Gated recurrent neural networks discover attention" (Zucchet et al., arXiv:2309.01775) proves a clean correspondence: gated linear RNNs with GLUs can express linearized self-attention exactly. [arxiv](https://arxiv.org/pdf/2309.01775)

(c) Extended LSTMs. Beck, Pöppel, Spanring, Auer, Prudnikova, Kopp, Klambauer, Brandstetter, Hochreiter, "xLSTM: Extended Long Short-Term Memory" (NeurIPS 2024 spotlight). Two new blocks: sLSTM with exponential gating and stabilization, and mLSTM with matrix memory and covariance update rule, fully parallelizable. [arXiv](https://arxiv.org/abs/2405.04517) Trained on 15B and 300B tokens of SlimPajama, compared to Llama and GPT-3 reproductions. xLSTM-7B (Beck et al. 2025) is the inference-efficient scaled model.

(d) Recursive Transformers. Universal Transformer (Dehghani, Gouws, Vinyals, Uszkoreit, Kaiser, ICLR 2019) applies the same Transformer block repeatedly with Adaptive Computation Time (Graves, arXiv:1603.08983) selecting per-token halting probability. [arXiv](https://arxiv.org/html/1807.03819v3) Relaxed Recursive Transformers (Bae et al., arXiv:2410.20672) add static LoRA modules per depth to recover capacity lost to weight-tying. **Mixture-of-Recursions** (Bae, Mila, Google DeepMind et al., arXiv:2507.10524, July 2025) — the closest current relative to what Dylan is building — combines parameter sharing across recursive depth with a Mixture-of-Experts-style router that selects per-token recursion depth, giving both parameter and compute adaptivity, and reports a different compute-optimal regime than Chinchilla, with parameter count contributing more than tokens. [Medium](https://medium.com/@akdemir_bahadir/mixture-of-recursions-mor-advancing-transformer-efficiency-through-parameter-sharing-and-1dcaa77881ac)

(e) Implicit / continuous models. **Neural ODEs** (Chen, Rubanova, Bettencourt, Duvenaud, NeurIPS 2018): the network is $\dot h(t) = f_\theta(h(t), t)$, depth becomes continuous, memory cost is $O(1)$ via the adjoint method. [University of Maryland](https://user.eng.umd.edu/~austin/ence688p.d/papers-reading/NN-Ordinary-Differential-Equations2018.pdf) **Deep Equilibrium Models** (Bai, Kolter, Koltun, NeurIPS 2019): the network *is* the fixed point $h^* = f_\theta(h^*, x)$, found by root-finding (Anderson acceleration, Broyden), differentiated implicitly, trained in constant memory regardless of effective depth. [arXiv](https://arxiv.org/pdf/1909.01377) **Liquid Neural Networks** (Hasani, Lechner, Amini, Rus, AAAI 2021): continuous-time RNNs with input-dependent time constants, demonstrated on autonomous driving with small parameter counts.

(f) Recursive (tree-structured) neural networks — distinct from recurrent. Socher et al. (2011–2013) — networks composing over a parse tree. Largely subsumed by Transformers for NLP, but conceptually relevant to MoR's tree-of-computation framing.

Neural Turing Machines (Graves, Wayne, Danihelka, arXiv:1410.5401, 2014) and **Differentiable Neural Computers** (Graves et al., *Nature* 2016) gave explicit differentiable external memory — the cleanest realization of a von-Neumann architecture inside a neural network. Their failure to scale is instructive: location-based addressing didn't generalize, and attention turned out to be the right primitive for content-based memory.

Benchmark efficiency claims, the honest version. Replicable wins: Griffin at 14B matching Llama-2 on downstream tasks with >6× fewer training tokens (De et al. abstract: "Griffin matches the performance of Llama-2 despite being trained on over 6 times fewer tokens"). [arXiv](https://arxiv.org/abs/2402.19427) Mamba-3B competitive with Transformers ~2× its size on the Pile, with up to 5× inference throughput at long context. xLSTM showing competitive scaling laws against Llama and Mamba reproductions at up to ~7B (Beck et al. 2024). Replicable failures: pure SSMs and linear RNNs trail full attention on retrieval/copying — Park et al. (arXiv:2402.04248) showed Mamba (up to 2.8B) "fails in vector-valued MQAR" though it matches Transformers on standard regression ICL.

4. Geometric and physical recursion

Renormalization group. Wilson's RG (1971) defines a flow on the space of effective theories under coarse-graining; fixed points of this flow are scale-invariant (critical phenomena, conformal field theories). The connection to deep learning is more than analogy: Mehta and Schwab (arXiv:1410.3831, 2014) showed an exact mapping between variational RG and stacked RBMs; Roberts, Yaida and Hanin's *Principles of Deep Learning Theory* (Cambridge 2022) builds the perturbative theory of wide networks on RG-style flow equations. The deep prior: scale invariance is a fixed point.

Strange attractors and dynamical systems. Lorenz, Hénon, Rössler — asymptotic geometry of recursion. Orbits do not converge to a point but to a fractal with non-integer dimension. Basins of attraction partition state space by which attractor an initial condition flows to. Liquid networks and continuous-time RNNs *are* dynamical systems and inherit this geometry.

Holographic principle. 't Hooft and Susskind's bound; Maldacena's AdS/CFT (1997). The MERA tensor network (Vidal 2007) realizes this as recursive coarse-graining; Pastawski, Yoshida, Harlow, Preskill (2015, HaPPY codes) make the recursion-as-error-correcting-code structure explicit.

Sacred geometry, treated rigorously. Penrose tilings, quasi-crystals, Apollonian gaskets, and the modular group's action on the upper half-plane are real recursive structures with mathematical content. Most pop-culture "sacred geometry" claims (e.g., that the golden ratio appears in arbitrary anatomy or art) are statistically unsupported — see Markowsky, "Misconceptions about the Golden Ratio," *The College Mathematics Journal* 23(1), 1992. Use the rigorous side.

5. Recursion in philosophy and cognition

Chomsky vs. Everett. Hauser, Chomsky, Fitch (*Science* 298, 2002, "The Faculty of Language: What Is It, Who Has It, and How Did It Evolve?") argued recursion is the uniquely human component of the faculty of language. Daniel Everett (*Current Anthropology* 46(4), 2005, "Cultural Constraints on Grammar and Cognition in Pirahã") [Pressbooks](https://pressbooks.openedmb.ca/wordandsentencestructures/chapter/debating-universal-grammar/) claimed Pirahã lacks clausal recursion, undermining UG. Nevins, Pesetsky, Rodrigues (*Language* 85(2), 2009, "Pirahã Exceptionality: A Reassessment")

[Pressbooks](https://pressbooks.openedmb.ca/wordandsentencestructures/chapter/debating-universal-grammar/) pushed back hard. Current state: even if Pirahã surface grammar lacks recursion, Pirahã speakers can *learn* recursive languages, so the *capacity* survives [Pressbooks](https://pressbooks.openedmb.ca/wordandsentencestructures/chapter/debating-universal-grammar/) — Chomsky's claim is about FL (faculty of language), not about every grammar's surface use. The debate is empirically live but the cognitive-capacity claim is the load-bearing one.

Hofstadter's strange loops. *Gödel, Escher, Bach* (1979, Pulitzer 1980) and *I Am a Strange Loop* (2007). The philosophical claim: self-reference at sufficient representational power yields a *self* — the loop through symbolic levels that returns to itself is, in Hofstadter's view, what consciousness *is*. [Wikipedia](https://en.wikipedia.org/wiki/I_Am_a_Strange_Loop) The formal anchor is Gödel's diagonal lemma; the parallel to the Y combinator and to Kleene's second recursion theorem is direct.

Predictive processing and the free-energy principle. Friston (Friston 2005, 2010; *Nature Reviews Neuroscience* 11, 2010, "The Free-Energy Principle: A Unified Brain Theory?") frames the brain as a hierarchical generative model that recursively minimizes variational free energy. Predictions flow top-down, prediction errors bottom-up; the recursive exchange iterates [Get Therapy Birmingham](https://gettherapybirmingham.com/the-predictive-mind-karl-fristons-free-energy-principle-and-its-implications-for-consciousness/) to a fixed point — Bayesian inference implemented as a contraction-mapping recursion. The cognitive-science analog of a deep equilibrium model.

6. Efficient models at scale — the practical question

What "efficient" means, precisely. Five distinct axes, often conflated: (i) *parameter efficiency* — quality per weight; (ii) *training compute efficiency* — quality per FLOP, measured against the Chinchilla frontier; (iii) *inference compute efficiency* — FLOPs per generated token; (iv) *inference memory efficiency* — KV-cache size, scaling linearly in sequence length for Transformers, bounded for RNNs/SSMs; (v) *energy efficiency* — joules per token, dominated by memory bandwidth.

The Transformer's weakness is (iv) and the long-sequence end of (iii); its strength is parallelism during training and (ii) at scale. The recurrent comeback specifically targets (iv) and decoding latency — Griffin and Mamba reduce decoding latency and increase throughput because per-step state is bounded. [arXiv](https://arxiv.org/abs/2402.19427) This is the cleanest case for recurrence.

Where Chinchilla holds and where it breaks. Hoffmann et al.'s 20 tokens/parameter ratio has been replicated: Besiroglu, Erdil, Barnett and You (Epoch AI, arXiv:2404.10102) re-derived the parametric law and concluded their fitted model "implies an optimal ratio of around 20 tokens per parameter, consistent with Chinchilla's central result," while also noting that Hoffmann et al.

"report implausibly narrow confidence intervals — intervals this narrow would require over 600,000 experiments, while they likely only ran fewer than 500." Sardana, Portes, Doubov & Frankle (MosaicML/Databricks, ICML 2024, arXiv:2401.00448) showed "model quality continues to improve as we scale tokens per parameter to extreme ranges (up to 10,000)" for inference-cost-aware deployment, recommending that LLM developers expecting ~1B inference requests "should train models smaller and longer than Chinchilla-optimal." [ACM Digital Library](https://dl.acm.org/doi/10.5555/3692070.3693840) For recurrent/SSM models, scaling exponents are different in ways not yet definitively measured — Mixture-of-Recursions reports that parameter count helps more than tokens in its regime.

Compute moat vs. architecture argument. The compute-moat camp (Sutton's "Bitter Lesson") says architecture matters less than compute and data. The architecture camp points to Griffin matching Llama-2 with >6× fewer tokens, MoR's adaptive depth, FlashAttention's hardware-aware speedup, and Mamba's selective scan as evidence that architecture bends the curve. Both are true: compute dominates absolute capability, architecture sets the constant. For a parameter-count-sensitive product (edge inference, on-device, long-context), architecture *is* the lever.

7. Compression and shape

Kolmogorov complexity. $K(x)$ is the length of the shortest program (on a fixed universal Turing machine) that outputs x — uncomputable, but tight up to an additive constant under change of UTM (invariance theorem; Solomonoff 1964, Kolmogorov 1965, Chaitin 1966).

Solomonoff induction and AIXI. The universal prior $M(x) \propto \sum_{p: U(p) = x*} 2^{-|p|}$ dominates among lower-semicomputable predictors. Hutter's AIXI (2005, *Universal Artificial Intelligence*) couples Solomonoff induction with sequential decision theory to define a universal, uncomputable, RL agent — "maximally intelligent in a precise, technical sense" [Danmackinlay](https://danmackinlay.name/notebook/aixi.html) (Hutter 2005). The Hutter Prize for lossless text compression operationalizes this on enwik9.

Recursion *is* compression. A neural network's weight count is the description length of its program; a parameter-shared / recursive / weight-tied architecture has shorter description length for the same expressive depth. This is the precise sense in which Dylan's intuition is correct: if VRU expresses what an LSTM expresses at a fraction of the parameters, it is moving toward the Solomonoff frontier from the practical side.

Inductive bias. The architecture is the prior. Convolutions encode translation equivariance; Transformers encode permutation equivariance with positional encoding; SSMs encode a continuous-time linear dynamical prior; recursion encodes the prior that the same operation is appropriate at every depth. The right inductive bias is the one whose Kolmogorov complexity is closest to the data-generating process's. For sequence data with long-range dependencies and locally-similar update structure, a recursive/selective bias is principled.

Where 4/π could appear meaningfully in VRU / DPPU

Three principled possibilities, in decreasing rigor:

1. **As a normalization on rectified/half-wave activations.** If the recurrence's state update applies a nonlinearity equivalent to half-wave rectification of a sinusoidal projection, then $4/\pi$ is the *exact* fundamental-harmonic gain — multiplying by $4/\pi$ recovers unit gain in the fundamental, the relevant geometric scaling for a square-wave-like signal. This is the cleanest principled story; it ties directly to Fourier analysis and is defensible from first principles.

2. **As a Cauchy-mean-width / projection constant.** If hidden states are interpreted geometrically as convex bodies in feature space and the update averages over directions, Finch's identity (cube mean-square width $= 1 + 4/\pi$, arXiv:1110.0671) [arxiv](https://arxiv.org/pdf/1110.0671) suggests $4/\pi$ as a natural anisotropy correction. More speculative; requires deriving the connection.

3. **As an empirical constant from a $\sin/\cos$ feature map.** If $\phi(x)$ uses circular/Fourier features, $4/\pi$ may appear as the angular-average constant in the kernel. Plausible; verify by deriving the kernel.

What it is *not*: it is not the golden ratio's geometric cousin, it is not the sharp isoperimetric constant, it is not a sharp constant in random matrix theory, and it does not make VRU automatically efficient. Whether $4/\pi$ is load-bearing or decoration depends on whether the architecture's derivation forces it.

Recommendations

Stage 1 — Frame the architecture in the existing taxonomy (this week).
- Position VRU/DPPU explicitly as a member of one of the five families above (most likely selective-SSM-adjacent or gated-linear-RNN-adjacent). The panel will ask which it is; have the answer.
- Write the recurrence in Katharopoulos / Dao–Gu duality form: $S_t = A_t S_{t-1} + B_t x_t$, $y_t = C_t S_t$. If the architecture *cannot* be cast this way, that itself is a major differentiation claim and must be defended explicitly.
- Derive *where* $4/\pi$ enters the update equations. The panel will be much more receptive to "$4/\pi$ is the fundamental-harmonic gain of our half-rectified projection" than to "$4/\pi$ is geometrically meaningful." Use the symbol $\kappa = 4/\pi$, not ϕ, to avoid collision with the golden ratio.

Stage 2 — Run the canonical benchmarks against named baselines (1–4 weeks).
- *Head-to-head required*: Mamba-2 at matched parameter count on the Pile; xLSTM at matched parameter count; Llama-2 7B as the Transformer reference.
- *Tasks required*: (i) WikiText-103 / Pile perplexity; (ii) Long Range Arena, especially Path-X; (iii) MQAR or passkey retrieval — *the* known weakness of recurrence per Park et al.; if VRU

survives here it is a major result; (iv) inference latency and throughput at sequence lengths 1k, 8k, 32k, 128k. Griffin's plot of held-out loss vs. training FLOPs is the gold-standard format.
- *Sanity check*: fit the scaling law $L(N, D) = E + A N^{-\alpha} + B D^{-\beta}$ on at least 4 model sizes. If α, β differ meaningfully from Chinchilla's, that is a real finding worth a section.

Stage 3 — Stress-test the recurrent failure modes (4–8 weeks).
- In-context learning past 3B — particularly *vector-valued* MQAR, the specific Mamba failure mode per Park et al. (arXiv:2402.04248).
- Length extrapolation — Griffin's strongest claim; [arXiv](https://arxiv.org/abs/2402.19427) VRU should match or beat it.
- Copying and exact retrieval — pure RNNs/SSMs lose; if VRU loses less, that is the strongest single result.

Stage 4 — Engage with Mixture-of-Recursions explicitly (8–12 weeks).
- Bae et al. 2025 (arXiv:2507.10524) is the closest competitor and the most-cited recent paper in adaptive-depth recursion. Either VRU is competitive on its benchmarks, or VRU is composable with MoR's router. Either is publishable.

Thresholds that should change the strategy.
- *Continue if*: VRU matches Mamba-2 perplexity at ≥50% fewer parameters on the Pile at ≥1B scale, *and* stays within 10% of a Transformer baseline on a retrieval task.
- *Pivot to hybrid if*: VRU underperforms on retrieval by >25% at scale. Add local attention layers — the panel will recognize this as the Griffin/Jamba/Zamba move, a known good answer.
- *Stop and rethink if*: $4/\pi$ does not have a derivable role and is only an empirical hyperparameter. Replace it with a learned scalar; the paper is stronger.

Caveats

- **The recurrent comeback is not yet a frontier-quality result.** No pure-recurrence or pure-SSM model has matched the best Transformers at >70B; Griffin is the strongest counterexample and it is a hybrid. Be calibrated to the panel about this.
- **Many recent claims are from blog posts and Medium articles, not peer-reviewed work.** "MoR is the Transformer killer" is a Medium framing; the paper itself (Bae et al. 2025) is more measured. The Griffin paper's claim of matching Llama-2 with >6× fewer tokens is rigorous; downstream summaries sometimes overstate. The Hawk paper claim is specifically that "Hawk exceeds the reported performance of Mamba on downstream tasks" [arXiv](https://arxiv.org/abs/2402.19427) with no "trained on half as many tokens" qualifier — earlier-circulated paraphrases that include that qualifier are not supported by the abstract.
- **The Chinchilla scaling law replication is nuanced.** Besiroglu et al.'s Epoch AI re-derivation confirmed the central ~20:1 ratio while flagging implausibly narrow original confidence intervals. The MosaicML/Sardana et al. work argues for substantially higher ratios under inference-cost-aware optimization. Do not cite Chinchilla as if it were a settled hard constant; cite it as a robust central estimate with known caveats.

- **The Pirahã debate is empirically contested.** Use it as an illustration of how the recursion-as-cognitive-universal claim is debated, not as settled fact.
- **The κ = 4/π claim must survive derivation.** The constant's reality in mathematics (square-wave fundamental, Brouncker, Cauchy mean-width of the cube) is not in question; its meaningfulness *in VRU* depends on whether it falls out of the math or is fitted.
- **Symbol hygiene.** Conflating φ (golden ratio ≈ 1.618) with $4/\pi$ (≈ 1.273) by using the symbol ϕ for both is a presentation failure waiting to happen. Use $\kappa = 4/\pi$ throughout.
- **"Strong benchmark results at a fraction of LSTM's parameter count" is a low bar.** LSTMs are a 1997 architecture; every architecture in §3 above has beaten them on parameter efficiency. The relevant comparisons are Mamba-2, xLSTM, and Griffin at matched parameters. Frame the result against the current frontier, not the historical one, or the panel will discount it on sight.
- **Park et al.'s failure-mode result is specific, not general.** They showed Mamba "fails in vector-valued MQAR" (a *non-standard retrieval* function) while matching Transformers on standard regression ICL — so the recurrent-models-can't-do-ICL claim must be qualified, not used as a blanket dismissal in either direction.